

Tutorial on Probability Theory

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Exercise 1.

True or False. Determine whether each statement is true or false, and explain your reasoning.

- (a) If a fair coin is tossed many times and the last eight tosses are all heads, then the chance that the next toss will be heads is somewhat less than 50%.
- (b) Drawing a face card (jack, queen, or king) and drawing a red card from a full deck of playing cards are mutually exclusive events.
- (c) Drawing a face card and drawing an ace from a full deck of playing cards are mutually exclusive events.

Exercise 2.

Four games, one winner. Below are four versions of the same game. Your archnemesis picks the version of the game, and then you choose how many times to flip a coin: 10 times or 100 times. Identify how many coin flips you should choose for each version of the game. It costs \$1 to play each game. Explain your reasoning.

- (a) If the proportion of heads is larger than 0.60, you win \$1.
- (b) If the proportion of heads is larger than 0.40, you win \$1.
- (c) If the proportion of heads is between 0.40 and 0.60, you win \$1.
- (d) If the proportion of heads is smaller than 0.30, you win \$1.

Exercise 3.

Poverty and language. The American Community Survey is an ongoing survey that provides data every year to give communities the current information they need to plan investments and services. The 2010 American Community Survey estimates that 14.6% of Americans live below the poverty line, 20.7%

speak a language other than English (foreign language) at home, and 4.2% fall into both categories.¹

- (a) Are living below the poverty line and speaking a foreign language at home disjoint?
- (b) What percent of Americans live below the poverty line and only speak English at home?
- (c) What percent of Americans live below the poverty line *or* speak a foreign language at home?
- (d) What percent of Americans live above the poverty line and only speak English at home?
- (e) Is the event that someone lives below the poverty line independent of the event that the person speaks a foreign language at home?

Exercise 4.

Guessing on an exam. In a multiple choice exam, there are 5 questions and 4 choices for each question (a, b, c, d). Nancy has not studied for the exam at all and decides to randomly guess the answers. What is the probability that:

- (a) the first question she gets right is the 5th question?
- (b) she gets all of the questions right?
- (c) she gets at least one question right?

Exercise 5.

Global warming. A Pew Research poll asked 1,306 Americans, “From what you’ve read and heard, is there solid evidence that the average temperature on earth has been getting warmer over the past few decades, or not?” The table below shows the distribution of responses by party and ideology, where the counts have been replaced with relative frequencies.

		<i>Response</i>			Total
		Earth is warming	Not warming	Don't Know Refuse	
<i>Party and Ideology</i>	Conservative Republican	0.11	0.20	0.02	0.33
	Mod/Lib Republican	0.06	0.06	0.01	0.13
	Mod/Cons Democrat	0.25	0.07	0.02	0.34
	Liberal Democrat	0.18	0.01	0.01	0.20
	Total	0.60	0.34	0.06	1.00

- (a) Are believing that the earth is warming and being a liberal Democrat mutually exclusive?
- (b) What is the probability that a randomly chosen respondent believes the earth is warming *or* is a liberal Democrat?
- (c) What is the probability that a randomly chosen respondent believes the earth is warming *given* that they are a liberal Democrat?

¹These estimates are from the 2010 American Community Survey.

- (d) What is the probability that a randomly chosen respondent believes the earth is warming *given* that they are a conservative Republican?
- (e) Does it appear that whether a respondent believes the earth is warming is independent of their party and ideology? Explain.
- (f) What is the probability that a randomly chosen respondent is a moderate/liberal Republican *given* that they do not believe the earth is warming?

Exercise 6.

Problem Set — homework

Work in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Unless stated otherwise, assume all events mentioned have strictly positive probability when conditioning on them.

1. Sum rule and complements.

- (a) Prove the *sum rule* for two events:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

(Hint: use the disjoint decomposition $A \cup B = (A \setminus B) \dot{\cup} B$.)

- (b) Deduce $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$ and the conditional version $\mathbb{P}(A \mid C) + \mathbb{P}(A^c \mid C) = 1$ when $\mathbb{P}(C) > 0$.

2. Union bounds and Bonferroni inequalities.

- (a) Using the sum rule, show the *union bound* (Boole's inequality) for two events:

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B).$$

- (b) Prove the sharp bounds

$$\max\{0, \mathbb{P}(A) + \mathbb{P}(B) - 1\} \leq \mathbb{P}(A \cap B) \leq \min\{\mathbb{P}(A), \mathbb{P}(B)\}.$$

(State clearly where each inequality comes from.)

3. Inclusion–exclusion via the sum rule.

- (a) Prove the three–event identity

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).$$

- (b) Show the conditional version given D with $\mathbb{P}(D) > 0$ by applying the identity to the conditional measure $\mathbb{P}(\cdot \mid D)$.

4. Law of total probability (LTP). Let $\{B_i\}_{i=1}^n$ be a partition of Ω with $\mathbb{P}(B_i) > 0$.

- (a) Prove the LTP:

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \mid B_i) \mathbb{P}(B_i).$$

- (b) Prove the conditional LTP: for $\mathbb{P}(C) > 0$,

$$\mathbb{P}(A \mid C) = \sum_{i=1}^n \mathbb{P}(A \mid B_i \cap C) \mathbb{P}(B_i \mid C).$$

5. **Bayes' theorem from definitions.** For events A and B with positive probability, starting only from $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$, derive Bayes' theorem:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \mathbb{P}(A)}{\mathbb{P}(B)}.$$

Then combine with the LTP to obtain the *Bayes' rule with a partition* $\{B_i\}$.

6. **Monotonicity and set difference.**

- (a) Prove: if $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$ and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$.
 (b) Show the conditional version: if $A \subseteq B$ and $\mathbb{P}(C) > 0$ then $\mathbb{P}(A \mid C) \leq \mathbb{P}(B \mid C)$ and $\mathbb{P}(B \setminus A \mid C) = \mathbb{P}(B \mid C) - \mathbb{P}(A \mid C)$.

7. **Independence and (non-)converses.**

- (a) Assume A and B are independent. Prove that $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.
 (b) Give a counterexample showing that $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ does *not* imply independence if $\mathbb{P}(B) = 0$ or 1.
 (c) (Verification) True or false? If $\mathbb{P}(A \mid B) > \mathbb{P}(A)$ and $\mathbb{P}(A), \mathbb{P}(B) \in (0, 1)$, then $\mathbb{P}(B \mid A) > \mathbb{P}(B)$. Prove your claim using Bayes' theorem.

8. **Conditioning distributes over unions and intersections.**

- (a) Prove the conditional sum rule:

$$\mathbb{P}(A \cup B \mid C) = \mathbb{P}(A \mid C) + \mathbb{P}(B \mid C) - \mathbb{P}(A \cap B \mid C),$$

assuming $\mathbb{P}(C) > 0$.

- (b) Deduce the conditional union bound $\mathbb{P}(A \cup B \mid C) \leq \mathbb{P}(A \mid C) + \mathbb{P}(B \mid C)$.

9. **Statement verification (justify each).** For arbitrary events A, B, C with $\mathbb{P}(C) > 0$, determine whether each statement is true or false, and give a proof or counterexample.

- (a) If $\mathbb{P}(A \cap B) = 0$ then $\mathbb{P}(A \mid C) + \mathbb{P}(B \mid C) \leq 1$.
 (b) If $\mathbb{P}(A \mid C) = \mathbb{P}(A)$ and $\mathbb{P}(B \mid C) = \mathbb{P}(B)$, then $\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \cap B)$.

(c) If $A \subseteq B$ then $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A)}{\mathbb{P}(B)}$.

10. **Challenge (optional): pairwise vs. mutual independence.** Construct events A, B, C on a finite sample space such that any two are independent (pairwise independence), but A, B, C are not mutually independent. Verify your construction by explicit probability calculations using the sum rule and conditional probabilities.